Active sensing explains cortical variability and improves visual information

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What looks like noise may reflect active sensing

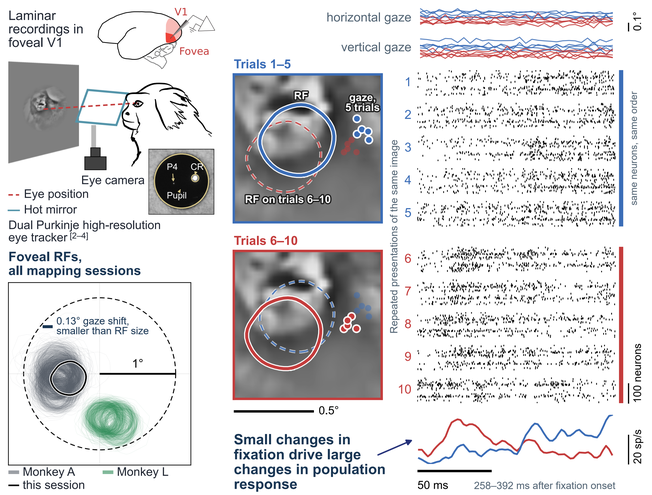
Neurons respond differently across repeated presentations of the same stimulus. These trial-to-trial differences are often described as "noise," but they can include structured responses arising from the animal's active sampling of the scene.

Even during fixation, small eye movements continually reshape visual input.

How much cortical variability comes from the eye?

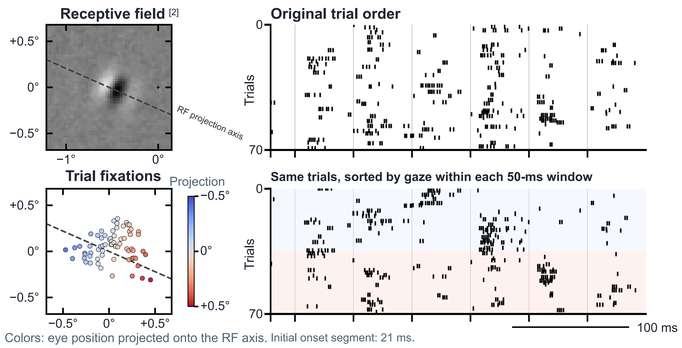
Do fixational eye movements help encode spatial detail?

1 Tiny gaze shifts reshape population responses



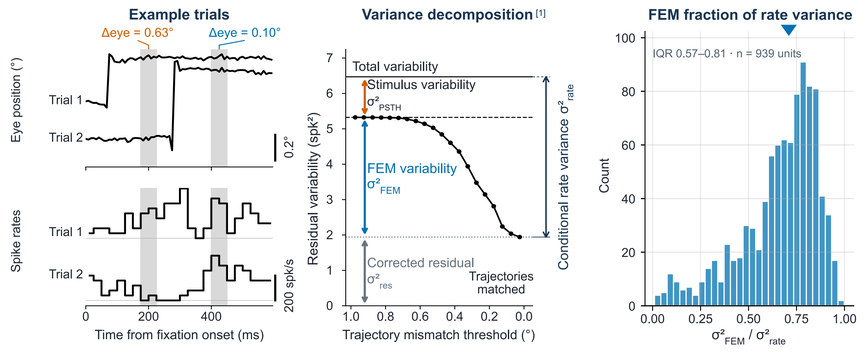
2	1,022	19	20 Hz
awake marmosets	foveal V1 units	recording sessions	repeated natural images

2 One neuron's "noise" tracks the eye

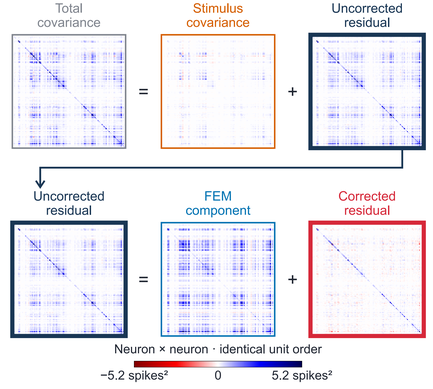


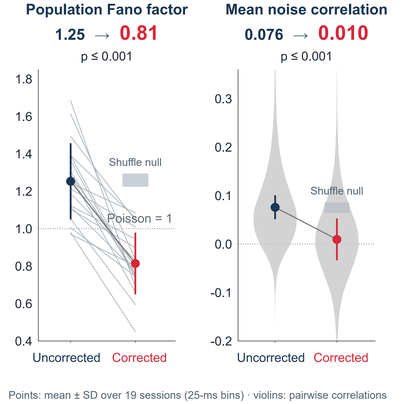
Colors: eye position projected onto the RF axis. Initial onset segment: 21 ms.

3 Eye movements drive most single-cell rate variance (71%)



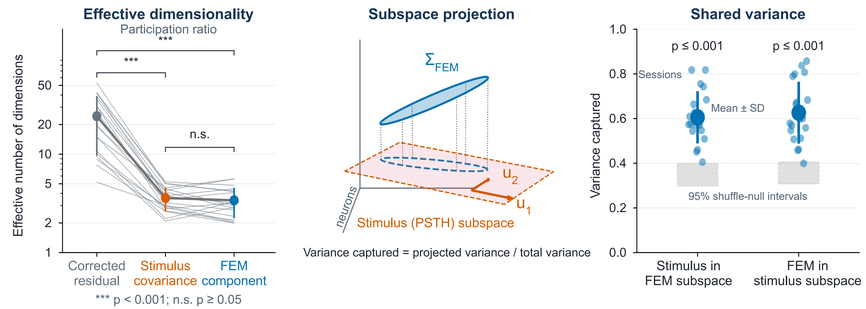
4 Correcting for gaze removes shared noise





Points: mean ± SD over 19 sessions (25-ms bins) · violins: pairwise correlations
Tests: correction vs shuffle (95% null) · corrected correlation 95% CI −0.010 to 0.029

5 Eye-driven variability follows stimulus dimensions

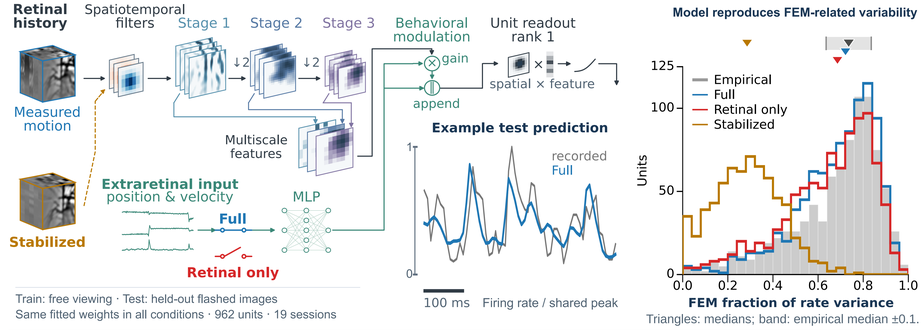


Variance captured = projected variance / total variance

*** p < 0.001; n.s. p ≥ 0.05

25-ms counts; alignment uses four leading dimensions per component.

6 Gaze-dependent retinal input predicts single-trial responses



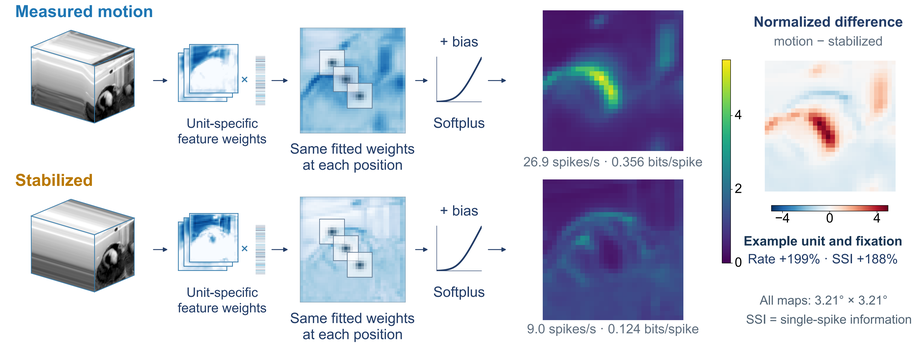
Train: free viewing · Test: held-out flashed images
Same fitted weights in all conditions · 962 units · 19 sessions

Triangles: medians; band: empirical median ±0.1.

Single-trial prediction performance
Median fraction of conditional rate variance explained

Stimulus-locked average (PSTH)	Full	Retinal only	Stabilized
0.139	0.235	0.226	0.042

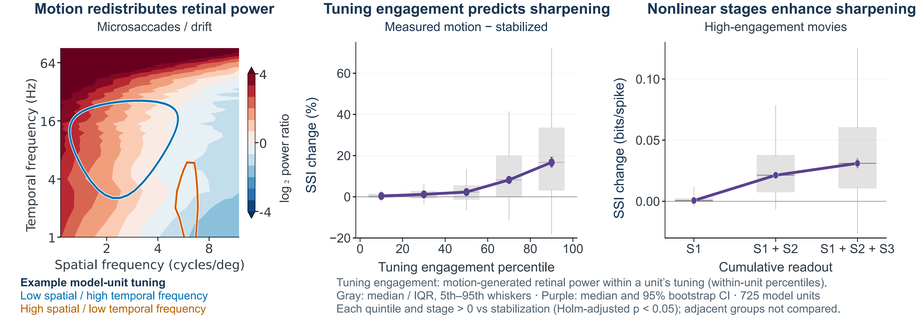
7 In the model, eye motion sharpens spatial coding



For a camera, motion means **blur.**
For vision, it can mean **detail.**

8 In the model, sharpening increases with tuning engagement

Eye motion converts spatial structure into temporal signals that engage neuronal tuning [5,6]



Tuning engagement: motion-generated retinal power within a unit's tuning (within-unit percentiles).
Gray: median / IQR, 5th–95th whiskers · Purple: median and 95% bootstrap CI · 725 model units
Each quintile and stage > 0 vs stabilization (Holm-adjusted p < 0.05); adjacent groups not compared.

Active sensing continues throughout fixation.

References
[1] McFarland 2016 · [2] Yates 2023 · [3] Wu 2023
[4] Ressmeyer 2026 · [5] Rucci & Victor 2015 · [6] Kuang 2012

Funding
NEI K99-EY037839-01 (D.P.R.); R01-EY030998 (J.F.M.); R01-EY018363 (M.R.).
Fuster Endowment (A.C.H.); NEI K99/R00-EY032179 (J.L.Y.); NIH/NEI T32-EY007043 (T.K.).

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Acknowledgements
Additional preprint coauthors: Dekel Galor · Ruei-Jr Wu · Michele Rucci
We thank Yonatan Abrham (Mitchell Lab) for collecting some of the data used in this report.